I do not see a demonstrated replacement for the old tier that meets your accuracy and cost targets. There is one worthwhile structural test, described below. The cost coincidence does not establish that leaders eliminated old sources: their savings are 95–127 units, and removing your old tier gives 153 units—not the entire 133–165 range.

For Q1, I would distinguish the following possibilities.

1. Exact contraction without dense leg storage: possible; the arithmetic saving is limited. [Derived]

Write $A = QF,\ P = QG$, with q_i the i-th row of Q. Then

$$H = YQ^\top, \qquad Y_{i:} = \left[(q_iF) \circ (q_iG) \circ w_2\right] F^\top.$$

Evaluate this in row blocks. Neither dense leg needs to be stored, although its entries are still evaluated.

Including weighting, its FLOP count is

$$2n^2(2r-1) + 2n^2 + nr(2n-1) + n^2(2r-1),$$

or approximately $4r/n$ units: **1.499 at $r = 384$, 0.874 at $r = 224$**, for H alone. Thus this answers the literal $O(n^2r)$, "without forming A, P" question, but does not remove your bottleneck.

One exact audit is worthwhile: **all Q-confined contributions share the final right factor $Q^\top$**. Sum their $n \times r$ intermediates across contraction types and sources before applying it. Each avoided projection saves approximately r/n units. With ten sources at $r = 224$, avoiding nine projections saves **1.97 units/layer**. Thin residual terms require their own treatment.

First falsifier: compare one saved source's streamed contraction with its dense reference, and inspect whether V29 already combines output projections. This costs **under 4 units**. If already combined, this opportunity is exhausted. Nothing is lost except roundoff.

2. Moving the closure into latent coordinates does not eliminate the cubic information. [Derived]

For

$$\kappa_3 = T \times_1 Q \times_2 Q \times_3 Q, \qquad f(q) = T[q, q, q],$$

the required slices satisfy

$$D_{3,i} = f(q_i), \qquad D_{21,i:} = \tfrac{1}{3} \nabla f(q_i) Q^\top.$$

A Hermite reformulation therefore asks for evaluations of a cubic polynomial and its gradient. A generic symmetric cubic has

$$\binom{r+2}{3}$$

coefficients: **1,898,400 at** $r = 224$ and **9,511,040 at** $r = 384$. Changing polynomial basis does not remove them. Your recorded Tucker costs still apply.

Likewise, 99% leg-energy retention means 10% relative Frobenius error in a leg, not 2% contraction error. The observed effective rank near 100 does not justify using $r = 100$.

I know no general exact $O(nr^2)$ identity here. Conversely, flat product spectra are **not an arithmetic lower bound**, nor do they establish that $(A \circ P)F^\top$ lacks exploitable structure. Polynomial sketches also retain a variance–cost tradeoff; avoiding explicit tensor products does not establish the required accuracy. PMLR Wacker et al.

3. Ordinary self-averaging fails, but a transported trace component is testable. [Derived + conjecture]

Freeze upstream states, absorb the current gains, and express the resulting old tensor T in an orthonormal r-dimensional basis. Fresh weight rows in this basis satisfy $w_i \sim N(0, \sigma^2 I_r)$, with $\sigma^2 = 2/n$. Define

$$t_c = \sum_a T_{aac}, \qquad F^2 = \|T\|_F^2.$$

For $i \neq c$, Wick contraction gives

$$\begin{aligned} \mathbb{E}\, T[w_i, w_i, w_c] &= 0, \\ \mathbb{E}[T[w_i, w_i, w_c] \mid w_c] &= \sigma^2 t^\top w_c, \\ \mathbb{E}\, T[w_i, w_i, w_c]^2 &= \sigma^6 \left(\|t\|^2 + 2F^2\right). \end{aligned}$$

Thus unconditional averaging deletes the signal. Even conditioning on the third row leaves variance $2\sigma^6 F^2$. Since

$$\|t\|^2 \leq (r+2)F^2/3,$$

that predictor's relative RMS error is at least

$$\sqrt{\frac{6}{r+8}},$$

or **12.4% at** $r = 384$. Conditioning additionally on actual means or other weight-dependent information is a different model; this calculation does not exclude every conditional closure.

The useful alternative retains the **actual quadratic weight fluctuations**:

$$T_{\mathrm{tr}} = \frac{\mathrm{sym}_3(I \otimes t)}{r+2}, \qquad T = T_{\mathrm{tr}} + H, \quad \mathrm{Tr}\, H = 0,$$

where sym_3 denotes the three-term sum. Its exact query is

$$T_{\mathrm{tr}}[w_i, w_i, w_c] = \frac{\|w_i\|^2 (t^\top w_c) + 2(w_i^\top w_c)(t^\top w_i)}{r+2}.$$

The expected relative squared residual is

$$\frac{2\|H\|_F^2}{2\|T\|_F^2 + \|t\|^2}.$$

Flat Hadamard spectra do not settle this quantity.

A memory-bearing generalization carries

$$T_{\text{old}} \approx \sum_{a=1}^{k} \operatorname{sym}_3(G_a \otimes v_a).$$

Under transport B,

$$G_a \leftarrow B G_a B^\top, \qquad v_a \leftarrow B v_a,$$

and

$$D_{21,ic} = \sum_a \big((G_a)_{ii} (v_a)_c + 2 (G_a)_{ic} (v_a)_i \big).$$

Transport within this family is exact. **The conjecture is that few components approximate old sources and incoming sources sufficiently well.** Their omitted residual is the loss.

A physical dense metric costs approximately **2 units/component/layer**. In your shared latent basis, metric transport plus reconstruction costs

$$2(r/n)^3 + (r/n)^2 + r/n,$$

or **0.621 units at** $r = 384$ and **0.288 at** $r = 224$, per component, plus shared basis maintenance. Fitting a new source to a fixed metric requires approximately $6r^2 n$ FLOPs (**0.422 units at** $r = 384$), plus the small solve. Basis maintenance and fitting cannot be called free.

This differs from slice-supported closure because it retains off-slice entries and independently transported historical state. It differs from dead item 3 **only if that experiment did not already carry this signed vector–metric coupling**. If item 4 established the best unrestricted low-CP-rank approximation of the old tensor, it also constrains this family; truncating existing hub columns is not equivalent.

First falsifier: the query experiment under Q3.

4. Earlier slices are observations, not generally a closed state. [Derived]

Exact transport requires

$$D^{+}_{21,ic} = \sum_{abd} B_{ia} B_{ib} B_{cd} T_{abd}, \qquad t^{+} = B[(B^{\top} B) : T].$$

A tensor supported on permutations of three distinct coordinate indices has zero current D_3, D_{21}, but generally nonzero next D_{21}. Covariance does not determine those missing entries.

For unrestricted symmetric tensors, supporting every possible future transport requires the information in $\binom{r+2}{3}$ coefficients. This is **not** a lower bound for your restricted source family or fixed remaining network. Historical slices could identify more of the tensor, but reconstruction does not supply a cheap update.

The minimum state for **2% error is undetermined by the supplied measurements**. Any claimed answer needs an approximation guarantee in the future-query metric, rather than a leg-energy threshold.

5. Evaluate fewer nonlinear query rows. [Conjecture; modest prospects]

Keep the existing factors and transport, but evaluate old-source slices only on rows where the actual closure coefficients are appreciable. Gaussian ReLU coefficients involving second and higher derivatives decay away from the threshold.

With $k = \alpha n$ retained rows, three reconstructed legs, four contractions, and one shared output projection cost approximately

$$(7S + 1)\alpha r/n$$

units/layer, plus basis maintenance and thin/elementwise work. For $S = 10, r = 224$, this is 15.53α **units/layer**.

This loses omitted rows' closure contributions. It differs from dead item 12 only if rows are skipped **before** contraction; if that was already tested, discard it.

First falsifier: on saved closure terms, measure the retained weighted energy versus row fraction. This costs **under 0.1 unit** for a small threshold sweep. If retaining roughly 10% of rows cannot meet the error allowance, screening cannot make the old tier nearly free. A successful screen still needs an end-to-end replay; small local error is insufficient.

For **Q2**, my ranking is speculative:

Rank	Hypothesis	Distinguishing signature
1	They retain historical content but share more contraction/output work.	The audit in item 1 finds redundant projections. Exact changes preserve the output; they cannot alone explain better raw MSE.
2	They carry one or a few historical vector–metric components.	Q3 finds a small aggregate residual; cost grows mainly with component count rather than source count.

3	They retain factors but evaluate only sensitive closure rows.	Cost tracks the fraction of neurons near the ReLU threshold; error rises sharply as that fraction is reduced.

Their costs and first falsifiers are given above. **None currently explains both the lower compute and lower raw MSE.** Compression does not automatically improve the closure's existing bias.

For **Q3**, my single next measurement would be **the aggregate old tensor's trace-component residual under actual and resampled transport queries**.

At layer 13 or 14:

1. Freeze the gains and old-source state. Absorb gains and orthonormalize the resulting old subspace.
2. Extract t directly from source factors. For a term $\mathrm{sym}_3(a, a, p)$, its contribution is

$$\|a\|^2 p + 2(a^\top p)a.$$

   This is roughly $16rn$ FLOPs/source including the analogous M terms: **0.003 units/source at $r = 384$.**
3. Use one panel of 32 actual next-layer rows and 16 independently resampled Gaussian panels. Compute exact source contributions and the transported trace prediction. **Sum sources before taking residual norms.**

Three leg projections and four contractions cost, per panel/source,

$$\frac{3mn(2n-1) + 4m^2(2n-1)}{2^{31}}, \qquad m = 32.$$

–

Seventeen panels across fourteen sources cost **23.23 units**, allowing an all-source denominator. Budget **under 35 units** including the old-basis transformation and trace calculations when using existing factors. That fits an hour at sustained throughput above approximately **21 MFLOP/s**; benchmark the first panel.

Report the aggregate residual divided by **full** D_{21} **RMS**, its actual-row value versus the resampled distribution, and aggregate residual energy versus summed per-source residual energy.

A residual above 2% rejects this trace replacement at that layer; it does not reject every possible metric. A small residual warrants testing several successive actual transports. This measures precisely the gap your current spectra leave unresolved: whether the expensive old content remains essential **after contraction and aggregation**.